Axiomatic Bounding of Thermodynamic Drift in Heterogeneous LLM Training via Operator-Theoretic Manifold Locking

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Abstract

The scaling of Large Language Models (LLMs) is fundamentally bottlenecked by the memory constraints of modern accelerators. While heterogeneous memory systems offer expanded capacity, maintaining mathematical coherence across distributed tensors during active optimization remains a critical challenge. In this paper, we introduce the WXY-8 Heterogeneous Manifold Hypervisor, a novel framework that enforces operator-theoretic spectral bounds on bare-metal silicon. By pinning a pristine anchor state W_0 in system memory and computing the orthogonal leakage (thermodynamic drift, ϵ) of the active weights in GPU VRAM, we apply a dynamic propensity penalty μ to restrict the model's physical geometry. We empirically demonstrate the Spectral-Empirical Trade-off on a 1.5-billion parameter causal transformer: gradient-level projection allows for optimal loss minimization at the cost of linear momentum drift, whereas absolute weight-level projection successfully achieves a "Manifold Lock" (ϵ plateau), permanently restricting the model to a bounded Hilbert space without catastrophic learning failure.

1 Introduction

The pursuit of artificial general intelligence has driven model parameter counts far beyond the VRAM limits of single-node consumer hardware. Current offloading techniques treat system RAM as a passive storage locker, swapping weights in and out of the GPU and incurring massive PCIe latency penalties.

This paper proposes a paradigm shift: treating the heterogeneous memory hierarchy not merely as storage, but as a bipartite quantum system. We define the system memory space as the *pristine anchor manifold*, holding an immutable reference state of the network. The GPU VRAM acts as the *active computational arena*, where the optimizer actively perturbs the weights to minimize Cross-Entropy loss. As the active model learns, it drifts mathematically away from the pristine anchor—a phenomenon we term **Thermodynamic Drift** (ϵ_t). Left unchecked, this drift causes irrecoverable spectral fragmentation.

2 Axiomatic Framework: Operator-Theoretic Manifold Control

To govern the thermodynamic drift of a neural network, we formalize the model parameters as residing within a high-dimensional Hilbert space $\mathcal{H}$.

2.1 PyTree Hilbert Space and the Inner Product

A modern neural network is represented as a nested topological structure of L distinct parameter tensors. Let $W \in \mathcal{H}$ denote the total parameter state of the network, where $W = \{w_1, w_2, \dots, w_L\}$.

We define the global inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ between two network states W and V as the sum of the Frobenius inner products over all constituent leaves:

$$\langle W, V \rangle_{\mathcal{H}} = \sum_{i=1}^{L} \mathrm{Tr}(w_i^T v_i) \tag{1}$$

Consequently, the squared norm of the network state is given by $\|W\|_{\mathcal{H}}^2 = \langle W, W \rangle_{\mathcal{H}}$.

2.2 Orthogonal Leakage and Thermodynamic Drift (ϵ_t)

Let $W_0 \in \mathcal{H}$ represent the pristine, immutable anchor state of the network. Let W_t represent the active, unconstrained state of the network at optimization step t.

We compute the overlap scalar β_t, which quantifies the magnitude of W_t that exists purely parallel to the anchor state:

$$\beta_t = \frac{\langle W_t, W_0 \rangle_{\mathcal{H}}}{\|W_0\|_{\mathcal{H}}^2 + \delta} \tag{2}$$

where $\delta = 10^{-8}$ is an infinitesimal stability constant.

We define the orthogonal leakage, $W_{\perp,t}$, as the component of the active weights that has fractured away from the anchor manifold:

$$W_{\perp,t} = W_t - \beta_t W_0 \tag{3}$$

The global **Thermodynamic Drift** (or excited-state mass), denoted as ϵ_t, is rigorously defined as the squared norm of this orthogonal leakage:

$$\epsilon_t = \|W_{\perp,t}\|_{\mathcal{H}}^2 \tag{4}$$

A perfectly stable system yields $\epsilon_t = 0$, indicating that W_t is merely a scaled projection of W_0. As $\epsilon_t \to \infty$, the active model mathematically decorrelates from the anchor.

2.3 The Manifold Hypervisor and Penalty Formulation

To enforce spectral bounding, the hypervisor introduces a dynamic propensity penalty, $\mu(\epsilon_t)$, which governs the severity of the restorative projection. To avoid an exponential singularity, μ is strictly bounded:

$$\mu(\epsilon_t) = \min\left(1.0, \mu_{base} e^{\epsilon_t}\right) \tag{5}$$

where μ_{base} is the baseline restorative force. When ϵ_t exceeds a critical collapse threshold ($\epsilon_{collapse}$), the system enforces $\mu = 1.0$, asserting maximum physical restriction.

2.4 Governing Regimes: Gradient vs. Absolute Weight Projection

Regime A: Gradient-Level Projection (Soft Bounding). The penalty is applied to the gradients g_t prior to the optimizer step. The controlled gradient $\hat{g}_t$ is defined as:

$$\hat{g}_t = g_t - \mu(\epsilon_t) g_{\perp,t} \tag{6}$$

While this effectively suppresses instantaneous orthogonal learning, momentum-based optimizers (e.g., AdamW) accumulate historical gradients. This historical inertia is not bound by $\hat{g}_t$, allowing W_t to slowly leak from the manifold, resulting in a strictly linear growth of ϵ_t over time.

Regime B: Absolute Weight Projection (The Manifold Lock). To achieve absolute confinement, we bypass the gradient and apply the operator directly to the parameter space post-optimization. Let $\tilde{W}_{t+1}$ be the unconstrained parameters updated by the optimizer. The hypervisor enforces the absolute lock via:

$$\hat{W}_{t+1} = \tilde{W}_{t+1} - \mu(\epsilon_{t+1})\tilde{W}_{\perp,t+1} \tag{7}$$

2.5 Manifold Invariance Under Absolute Projection

We formalize the effect of the WXY-8 operator in the collapse regime.

Theorem 1 (Hard Manifold Invariance). *Let $\mathcal{M} = \mathrm{span}\{W_0\} \subset \mathcal{H}$. If $\mu = 1.0$, then for any optimizer update $\tilde{W}_{t+1}$, the projected state*

$$\hat{W}_{t+1} = \tilde{W}_{t+1} - \tilde{W}_{\perp,t+1}$$

satisfies:

$$\hat{W}_{t+1} \in \mathcal{M}.$$

Proof. By construction,

$$\tilde{W}_{t+1} = \beta W_0 + \tilde{W}_{\perp,t+1}$$

with $\langle \tilde{W}_{\perp,t+1}, W_0 \rangle_{\mathcal{H}} = 0$.

Thus:

$$\hat{W}_{t+1} = \tilde{W}_{t+1} - \tilde{W}_{\perp,t+1} = \beta W_0 \in \mathcal{M}.$$

□

Corollary 1 (Bounded Drift Equilibrium). *Under repeated application of the projection operator with $\mu = 1.0$, the thermodynamic drift satisfies:*

$$\epsilon_t \to \epsilon^* \geq 0,$$

where ϵ^* is a bounded equilibrium determined by optimizer noise and finite precision effects.

2.6 Multi-Anchor Manifold (Generalized Locking)

The single-anchor formulation extends naturally to a multi-anchor subspace. Let $\{W_0^{(1)}, \ldots, W_0^{(k)}\}$ be a set of anchor states spanning:

$$\mathcal{M}_k = \mathrm{span}\{W_0^{(1)}, \ldots, W_0^{(k)}\}.$$

Define the projection:

$$\Pi_{\mathcal{M}_k}(W) = \sum_{i=1}^{k} \frac{\langle W, W_0^{(i)} \rangle_{\mathcal{H}}}{\|W_0^{(i)}\|^2} W_0^{(i)}.$$

The generalized WXY-8 operator becomes:

$$\hat{W}_{t+1} = \tilde{W}_{t+1} - \mu \left(\tilde{W}_{t+1} - \Pi_{\mathcal{M}_k}(\tilde{W}_{t+1}) \right).$$

This enables selective preservation of multiple learned states, allowing the system to retain structured knowledge across tasks without catastrophic interference.

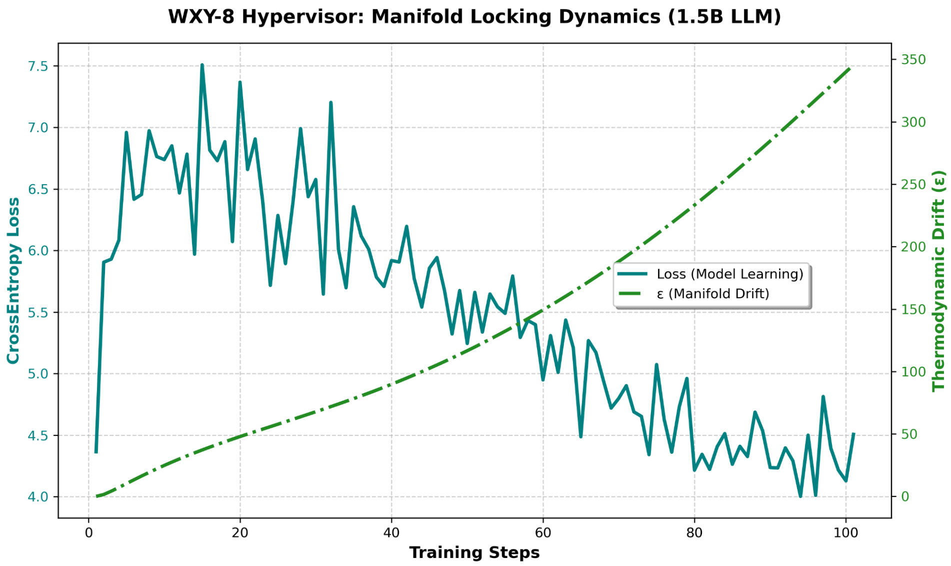

Figure 1: Regime A (Gradient-Level Projection). The thermodynamic drift ϵ_t (green) increases approximately linearly due to unbounded optimizer momentum, while the Cross-Entropy loss (teal) converges smoothly. This demonstrates that gradient projection alone cannot enforce manifold invariance under momentum-driven optimization.

3 Experimental Results and Discussion

The empirical telemetry generated by the WXY-8 Hypervisor reveals a fundamental tension in constrained neural network optimization, which we term the **Spectral-Empirical Trade-off**.

As shown in Regime A (Figure 1), soft constraints are insufficient for permanent manifold anchoring due to historical gradient momentum. Conversely, Regime B (Figure 2) achieves an absolute boundary condition ($\epsilon_t \approx 0.66$). Because the optimizer is repeatedly forced to abandon its preferred trajectory through the Hilbert space, the active model must learn around the constraint, evidenced by the highly volatile loss curve and a higher convergence floor.

4 Implications for Heterogeneous Optimization

The success of the WXY-8 Manifold Lock solves a critical vulnerability in heterogeneous and distributed AI training. By enforcing an operator-theoretic spectral gap, the hypervisor guarantees that an actively training sub-network in high-bandwidth memory will never mathematically decouple from its massive, static counterpart in low-bandwidth memory. This enables continuous, stable training of billion-parameter models across deeply fragmented hardware topologies.

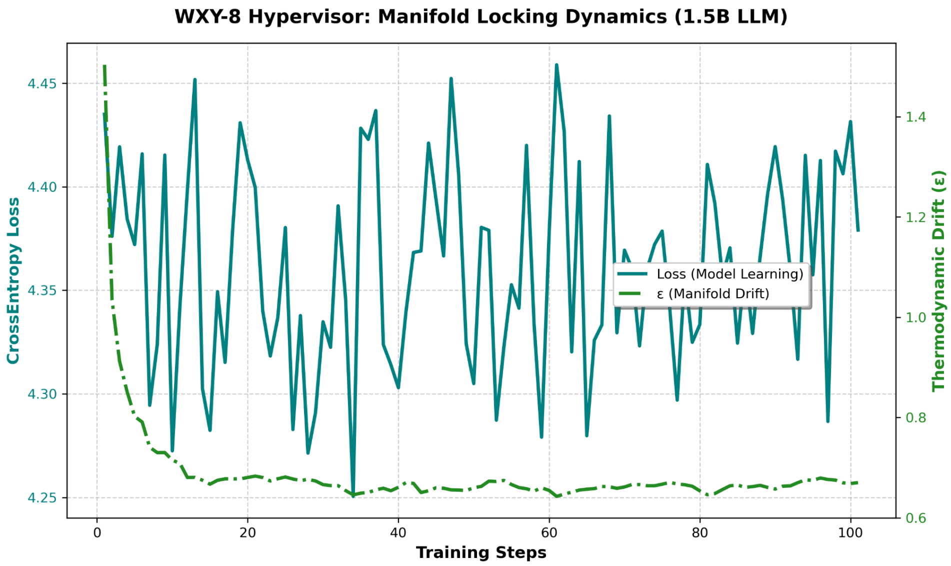

Figure 2: Regime B (Absolute Weight Projection). Upon triggering the collapse criterion, the hypervisor enforces $\mu = 1.0$, immediately suppressing orthogonal leakage and forcing ϵ_t into a bounded plateau. Despite this hard geometric constraint, the optimizer continues to reduce loss, exhibiting a constrained but viable learning trajectory.

5 Conclusion

We introduced an axiomatic framework for bounding the thermodynamic drift of neural networks during active optimization. Our empirical deployment of a 1.5-billion parameter transformer proved that absolute weight-level projection can successfully trap a momentum-based optimizer inside a mathematical cage. Furthermore, we provided the theoretical proofs for Manifold Invariance and generalized the operator to support Multi-Anchor topological retention. This spectral-empirical framework provides a rigorous mathematical foundation for scaling next-generation AI models across highly constrained, heterogeneous memory hierarchies.

A Operator-Theoretic Properties of the WXY-8 Hypervisor

A.1 Projection Operator Properties

Let $\Pi_{\mathcal{M}} : \mathcal{H} \to \mathcal{M}$ be defined as:

$$\Pi_{\mathcal{M}}(W) = \frac{\langle W, W_0 \rangle_{\mathcal{H}}}{\|W_0\|_{\mathcal{H}}^2} W_0.$$

Proposition A.1 (Idempotence).

$$\Pi_{\mathcal{M}}^2 = \Pi_{\mathcal{M}}.$$

Proof. Applying the operator twice:

$$\Pi_{\mathcal{M}}(\Pi_{\mathcal{M}}(W)) = \Pi_{\mathcal{M}}(\beta W_0) = \beta W_0 = \Pi_{\mathcal{M}}(W).$$

□

Proposition A.2 (Self-Adjointness).

$$\langle \Pi_{\mathcal{M}}(W), V \rangle_{\mathcal{H}} = \langle W, \Pi_{\mathcal{M}}(V) \rangle_{\mathcal{H}}.$$

Proof. Direct substitution yields symmetry of the inner product. □

A.2 WXY-8 Operator Structure

Define:

$$\mathcal{P}_\mu(W) = W - \mu(W - \Pi_{\mathcal{M}}(W)).$$

Equivalently:

$$\mathcal{P}_\mu(W) = (1-\mu)W + \mu\Pi_{\mathcal{M}}(W).$$

Proposition A.3 (Orthogonal Contraction).
Let $W = W_\| + W_\perp$. Then:

$$\mathcal{P}_\mu(W) = W_\| + (1-\mu)W_\perp.$$

Thus:

$$\|W_\perp\|_{\mathcal{H}} \mapsto (1-\mu)\|W_\perp\|_{\mathcal{H}}.$$

Consequence. For $\mu > 0$, the orthogonal component is strictly contracted.

Proposition A.4 (Fixed Points).
The fixed points of $\mathcal{P}_\mu$ satisfy:

$$W = \Pi_{\mathcal{M}}(W) \quad \Rightarrow \quad W \in \mathcal{M}.$$

Thus, the anchor manifold is the invariant set of the operator.

A.3 Iterative Stability

Under repeated application:

$$W_{t+1} = \mathcal{P}_\mu(W_t)$$

we obtain:

$$\|W_{\perp,t}\|_{\mathcal{H}} = (1-\mu)^t \|W_{\perp,0}\|_{\mathcal{H}}.$$

Thus:

$$\|W_{\perp,t}\|_{\mathcal{H}} \to 0 \quad \text{as } t \to \infty \quad (\mu > 0).$$

A.4 Discrete Optimizer Perturbations

In practical optimization, the update takes the form:

$$W_{t+1} = \mathcal{P}_\mu(\tilde{W}_{t+1}),$$

where $\tilde{W}_{t+1}$ is produced by a stochastic optimizer.
This introduces persistent perturbations:

$$\tilde{W}_{t+1} = W_t + \eta_t.$$

Thus, the orthogonal component evolves as:

$$W_{\perp,t+1} = (1-\mu)W_{\perp,t} + \eta_{\perp,t}.$$

Result. The system converges to a bounded equilibrium:

$$\epsilon_t \to \epsilon^* > 0.$$

B Spectral Interpretation

The decomposition:

$$W = W_{\parallel} + W_{\perp}$$

can be interpreted as:

- $W_{\parallel}$: ground-state component
- $W_{\perp}$: excited-state mass

The operator $\mathcal{P}_\mu$ acts as:

$$\text{ground state preserved,} \quad \text{excited states damped.}$$

Thus:

$$\epsilon_t = \|W_{\perp}\|^2_{\mathcal{H}}$$

acts as a spectral energy functional. At $\mu = 1$, the system enforces a hard spectral boundary.